

PARTS LIST

Semiconductors:

IC1	- NE555 timer
T1, T2	- BC547 npn transistor
D1, D2	- 1N4007 rectifier diode
BR1	- DB107 bridge rectifier
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R6	- 2.2-kilo-ohm
R2	- 10-kilo-ohm
R3	- 220-kilo-ohm
R4	- 1-kilo-ohm
R5	- 4.7-kilo-ohm
VR1	- 1-mega-ohm potmeter

Capacitors:

C1, C3	- 1000 μ F, 25V electrolytic
C2, C4	- 0.1 μ F ceramic disk
C5	- 0.01 μ F ceramic disk

Miscellaneous:

CON1-CON3	- 3-pin connector
X1	- 230V AC primary to 9V, 300mA secondary transformer
RL1	- 9V, 1C/O relay
	- PIR sensor module

Test Points

Test point	Details
TP0	GND
TP1	Around 9V
TP2	Around 3.3V after motion is detected
TP3	0-9V depending on human movement in front of the PIR
TP4	Around 9V after detecting human movement

transistor T2, and the appliance is switched on.

When capacitor C3 charges above two-thirds of the supply voltage, IC1's output pin 3 goes low and de-energises the relay and switches off the appliance after some delay that can be adjusted through potmeter VR1.

Whenever motion is detected by the PIR, its output pin goes high (around 3.3V) for a while depending on the setting on the PIR. The high signal from the PIR is fed to the base of transistor T1, which, in turn, discharges capacitor C3 through resistor R4. When the capacitor's charge (voltage) reaches less than two-thirds of the power supply, output pin 3 of IC1 goes high again (initial stage) and load is switched on.

When the load is switched off, LED2 glows. This indicates that the

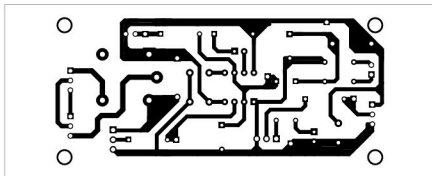


Fig. 3: Actual-size PCB pattern of PIR sensor based power saver

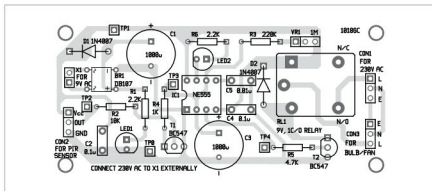


Fig. 4: Component layout of the PCB

circuit is under power-save mode.

Construction and testing

An actual-size, single-side PCB for the PIR sensor based power saver is shown in Fig. 3 and its component layout in Fig. 4. Enclose the PCB in a small box in such a way that you can easily connect 230V AC input to CON1 and the light/fan to CON3 at rear end of the box. Connect the PIR using a 3-wire cable to the PCB at CON2 and install it at a suitable location in your room.

Before using the PIR in the circuit, manually check it by connecting Vcc and GND pins of PIR to a 9V (or 12V) battery. Then check for change in voltage at signal output pin with respect to ground by waving your hand in front of the sensor. Adjust sensitivity and time controls of the PIR as per requirement (else turn both presets clock-wise to have highest sensitivity and high time signal). The dome surface of the PIR should be clean for better sensing.

After manually checking the PIR, remove the battery and connect the PIR to the PCB. You can assemble the circuit on the PCB with terminal connectors for CON1 (input) and CON3 (output).

Verify that voltages at the test points are as per the table before using the circuit. Check to see if LED2 is off and relay is energised. After some time, LED2 will glow and the relay will get de-energised. Your circuit is now ready!

EFY notes. Some PIR sensors require an initial stabilisation time of 10 to 60 seconds in order to function properly. During this time, any motion in its field-of-view (near 15m range) should be avoided.

Main DC voltage at TPI is unregulated. **EFY**



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